

MATERIAL SAFETY DATA SHEET

Mixed Polyol Material



MSDS DATE: 16/04/2021

FORMER DATE: --/--/----

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

PRODUCT NAME: Bio-glycol mixture
SYNONYM: Mixed polyol material
EC NUMBER: Not assigned
CAS NUMBER: Not assigned
REACH REGISTRATION NUMBER: No registration number is given for this substance since it is exempted from Registration

1.2 Relevant identified uses of the substance or mixture and uses advised against

IDENTIFIED USES: Product and process orientated research and development (PPORD)
APPLICATIONS ADVISED AGAINST: Only for product and process orientated research and development (PPORD)

1.3 Details of the supplier of the safety data sheet

SUPPLIER: UPM Biochemicals GmbH
ADDRESS: Am Haupttor, Building 4614, 06237 Leuna
E-MAIL ADDRESS: msdsBC@upm.com
TELEPHONE: +49 3461 519 5000

1.4 Emergency telephone number

EMERGENCY NUMBER: 112
Poison Information Centres (in Germany), open 24 h daily
Berlin (responsible for Berlin and Brandenburg) +49 030 19240
Bonn (responsible for North Rhine-Westphalia) +49 0228 19240
Erfurt (responsible for Mecklenburg-Western Pomerania, Saxony, Saxony-Anhalt and Thuringia) +49 0361 730 730
Freiburg (responsible for Baden-Württemberg) +49 0761 19240
Göttingen (responsible for Lower Saxony, Bremen, Hamburg and Schleswig-Holstein) +49 0551 19240
Homburg (responsible for Saarland) +49 06841 19240
Mainz (responsible for Rhineland-Palatinate and Hesse) +49 06131 19240
Munich (responsible for Bavaria) +49 089 1924

See section 16 for the list of emergency telephone numbers in other countries.

SECTION 2: HAZARD IDENTIFICATION

2.1 Classification of the substance or mixture

STOT RE 2 H373 (kidney) (Oral)

2.2 Label elements



GHS08

Signal word: Warning

Hazard Statements: H373: May cause damage to organs (oral) through prolonged or repeated exposure (oral)

Precautionary Statements: P260: Do not breathe fume/gas/mist/vapours.
P314: Get medical advice/attention if you feel unwell.
P501: Dispose of contents/container in accordance with local/regional regulations

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2.3 Other hazards

None

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

3.1 Substances

EC number:	CAS number:	Substance name:	Weight % range	Classification EC 1272/2008 (CLP):
203-473-3	107-21-1	Ethane-1,2-diol	15-30	Acute Tox. 4 * STOT RE 2

*) Harmonised classification - Annex VI of Regulation (EC) No 1272/2008 (CLP Regulation)

3.2. Mixtures

Not applicable

3.3 Other information

The substance is classified as target organ toxicant through repeated exposure because ethane-1,2-diol content in the product exceeds the generic concentration limit of ≥ 10 %.

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

EMERGENCY OVERVIEW: Treat symptomatically and supportively. Consult a physician. Show this safety data sheet to the doctor in attendance.

RESPIRATION: Remove the affected person to fresh air. If breathing has stopped, administer artificial respiration. Obtain medical assistance immediately.

SKIN CONTACT: Flush the contaminated skin with water. Remove contaminated clothing and shoes. Get medical aid if irritation develops and persists. Wash clothing before reuse.

EYE CONTACT: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Get medical aid.
Skin:

INGESTION: Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical aid.

4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in Section 11.

4.3 Indication of any immediate medical attention and special treatment needed

No data available

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.

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5.3 Advice for firefighters

In the case of fire, if applicable, use flame protective clothing and a respiratory device independent from ambient air. During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. For personal protection see section 8.

6.2 Environmental precautions

Do not let product to enter drains.

6.3 Methods and material for containment and cleaning up

Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container.

6.4 Reference to other sections

Treat the absorbed material according to Section 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep container tightly closed.

7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place.

7.3 Specific end use(s)

See subsection 1.2.

SECTION 8: EXPOSURE CONTROL/PERSONAL PROTECTION

8.1 Control parameters

Occupational exposure limits and/or biological limit values

Do not contain any components with workplace control parameters

8.2 Exposure controls

Appropriate engineering controls

Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower. Use adequate ventilation to keep airborne concentrations low. Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/ Face protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin protection

Wear appropriate protective clothing to prevent skin exposure

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Hand protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Environmental exposure controls

Any discharge of material to sewer needs to be avoided. Any residues, including spills and contaminants on the floor, are to be collected as hazardous waste in enclosed appropriate containers. The residues of liquid material are treated as hazardous waste. Closed sinks/basins or other appropriate containers shall be used to prevent discharge to waste-and/or surface water.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance	Liquid, viscous
- Physical state:	Colorless
- Color:	
Odor:	odourless (ethane-1,2-diol)
Specific density:	1.0273 g cm ⁻³
Odor Threshold:	No data available
pH value:	No data available
Melting point/Freezing point:	-60 °C (propane-1,2-diol); -13 °C (ethane-1,2-diol)
Initial boiling point/boiling range:	189 °C
Flash point:	99 °C (propane-1,2-diol); 111 °C (ethane-1,2-diol)
Evaporation rate:	No data available
Flammability (solid, gas) :	No data available
Vapor pressure:	0.08 mm Hg @ 20 °C (propane-1,2-diol); 0.0123 mm Hg @ 25 °C (ethane-1,2-diol)
Lower explosion limit	2.6 vol % (propane-1,2-diol)
Upper explosion limit	12.6 vol % (propane-1,2-diol)
Vapor density:	2.63 - (Air = 1.0) (propane-1,2-diol)
Solubility(ies) :	Miscible with water (propane-1,2-diol); 1 000g/L (ethane-1,2-diol)
Partition coefficient: n-octanol/water :	-1.07 (propane-1,2-diol); -1.36 (ethane-1,2-diol)
Auto-ignition temperature:	>400 (propane-1,2-diol); 398 °C (ethane-1,2-diol)
Decomposition temperature:	No data available
Viscosity:	43.4 mPa · s (propane-1,2-diol); 16.1 mPa · s (dynamic) (ethane-1,2-diol)
Oxidising properties:	No data available

9.2 Other information

No other information available

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

No dangerous reactivity under normal environmental conditions (air pressure 0.9 to 1.1 bar and temperature -20 °C to +40 °C). Hygroscopic: absorbs moisture or water from the air.

10.2 Chemical stability

Stable under recommended storage conditions.

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10.3 Possibility of hazardous reactions

No dangerous reactions are expected when properly used.

10.4 Conditions to avoid

Excess heat, moist air.

10.5 Incompatible materials

Oxidizing agents.

10.6 Hazardous decomposition products

Carbon monoxide, carbon dioxide, aldehydes

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

ACUTE TOXITY

LD₅₀ Oral - rat - 20.000 mg/kg (propane-1,2-diol); LD₅₀ Oral - rat - 7 712 mg/kg (ethane-1,2-diol)
LD₅₀ Dermal - rabbit - 20.800 mg/kg (propane-1,2-diol); LD₅₀ Dermal - mouse - 3 500 mg/kg (ethane-1,2-diol)

SKIN CORROSION/IRRITATION

Human, skin: Mild irritant (propane-1,2-diol), Rabbit, skin: not irritant (ethane-1,2-diol)

SERIOUS EYE DAMAGE/IRRITATION

Rabbit, eye: Mild irritant (Draize test) (propane-1,2-diol); Rabbit, skin: not irritant (ethane-1,2-diol)

RESPIRATORY OR SKIN SENSITISATION

Not sensitising (propane-1,2-diol); Not sensitising (ethane-1,2-diol)

GERM CELL MUTAGENICITY

Not mutagenic (propane-1,2-diol); Not mutagenic (ethane-1,2-diol)

CARSINOGENICITY

Not identified as probable, possible or confirmed human carcinogen by IARC (propane-1,2-diol); not considered to be classified as carcinogen (ethane-1,2-diol)

REPRODUCTIVE TOXICITY

Not a reproductive toxicant (propane-1,2-diol); Not a reproductive toxicant (ethane-1,2-diol)

STOT-SINGLE EXPOSURE

Not toxic after single exposure (propane-1,2-diol); Not toxic after single exposure (ethane-1,2-diol)

STOT-REPEATED EXPOSURE

Not toxic after repeated exposure (propane-1,2-diol); Not toxic after repeated exposure (ethane-1,2-diol)

ASPIRATION HAZARD

Aspiration hazard is not expected.

POTENTIAL HEALTH EFFECTS:

May cause reversible Gastrointestinal disturbance, Nausea, Headache, Vomiting, Central nervous system depression (propane-1,2-diol)

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SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

Short-term toxicity:

Invertebrates: 48-h EC50 (immobilisation; Daphnia magna): > 10 000 mg/L (propane-1,2-diol)
48-h NOEC (mortality; Daphnia magna): > 13 020 mg/L (propane-1,2-diol)
48-h EC50 (immobilisation; Daphnia magna): 13 900 mg/L (ethane-1,2-diol)

Fish: 96-h NOEC (mortality; Fathead minnow): 52 930 mg/L (propane-1,2-diol)
96-h NOEC (LC50; Pimephales promelas): 72 860 mg/L (ethane-1,2-diol)

12.2 Persistence and degradability

Readily biodegradable (propane-1,2-diol)
Readily biodegradable (ethane-1,2-diol)

12.3 Bioaccumulative potential

Based on the log Pow values of the constituents the substance has no bioaccumulation potential

12.4 Mobility in soil

Based on the result of the octanol/water partition coefficient of the constituents, the substance is expected to have a low mobility in soil.

12.5 Results of PBT and vPvB assessment

The substance is not PBT / vPvB.

12.6 Other adverse effects

No further information is available.

12.7 Additional information

None.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Waste should only be disposed of via a licensed waste contractor.

Treatment of contaminated packaging

A specific treatment is not required.

Waste codes

There is no EC waste code for this material according to the guideline 75/422/ECC.

Specific precautionary measures

None.

Relevant EC or other regulations

None.

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SECTION 14: TRANSPORT INFORMATION

14.1 UN number

Not regulated.
The material is not considered a hazardous good

14.2 UN proper shipping name

ADR/RID: Not dangerous goods
IMDG: Not dangerous goods
IATA: Not dangerous goods

14.3 Transport hazard class(es)

None.

14.4 Packing group

None

14.5 Environmental hazards

Not hazardous

14.6 Special precautions for user

See section 7.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

No data available

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1907/2006 (REACH)

15.2 Chemical safety assessment

In accordance with Regulation (EC) No. 1907/2006 (REACH) Article 14, Chemical Safety Assessments has been carried out for the components of the substance, propane-1,2-diol and ethane-1,2-diol.

SECTION 16: OTHER INFORMATION

16.1 Additions, Deletions, Revisions

Version 1.0.

This safety data sheet is drawn up to comply with the requirements Annex II of Regulation (EC) No. 1907/2006 (REACH), as amended by Annex I to Commission Regulation (EU) No. 453/2010 of 20 May 2010.

16.2 Key or legend to abbreviations and acronyms

- CLP - Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures
 - LD50: Lethal dose of the substance that causes 50 % mortality of the test population
- OSHA - Occupational Safety and Health Administration (USA)
- NOEC - No observed effect concentration
- PBT/vPvB - Persistent, bioaccumulative and toxic/ very persistent and very bioaccumulative
- REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
- STOT RE - Specific Target Organ Toxicity, Repeated Exposure
- STOT SE - Specific Target Organ Toxicity, Single Exposure

16.3 Emergency telephone number

Europe wide emergency number 112
USA, Canada, Brazil 911

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Poison information centres **AUSTRIA** (Vienna, Wien) +43 1 406 43 43; **BELGIUM** (Brussels, Bruxelles) +32 70 245 245; **BULGARIA** (Sofia) +359 2 9154 409; **CZECH REPUBLIC** (Prague, Praha) +420 224 919 293; **DENMARK** (Copenhagen) 82 12 12 12; **ESTONIA** (Tallinn) 112; **FINLAND** (Helsinki) +358 9 471 977; **FRANCE** (Paris) +33 1 40 0548 48; **GERMANY** (Berlin) +49 30 19240; **GREECE** (Athens, Athinai) +30 10 779 3777; **HUNGARY** (Budapest) 06 80 20 11 99; **ICELAND** (Reykjavik) +354 525 111 or +354 543 2222; **IRELAND** (Dublin) +353 1 8379964; **ITALY** (Rome) +3906 305 4343; **LATVIA** (Riga) +371 704 2468; **LITHUANIA** (Vilnius) +370 5 236 20 52 or +370 687 53378; **MALTA** (Valletta) 2425 0000; **NETHERLANDS** (Bilthoven) +31 30 274 88 88; **NORWAY** (Oslo) 22 591300; **POLAND** (Gdansk) +48 58301 65 16 or +48 58 349 2831; **PORTUGAL** (Lisbon, Lisboa) 808 250 143; **ROMANIA** (Bucharest) +40 21 3183606; **SLOVAKIA** (Bratislava) +421 2 54 77 4166; **SLOVENIA** (Ljubljana) + 386 41 650500; **SPAIN** (Barcelona) +34 93 227 98 33 or +34 93 227 54 00 bleep 190; **SWEDEN** (Stockholm) 112 or +46 833 12 31 (mon-fri 9.00-17.00); **UNITED KINGDOM** (London) 112 or 0845 4647 (NHS Direct).

NOTICE TO READER

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